



# **Patuakhali Science and Technology University**

## **Faculty of Computer Science and Engineering**

Patuakhali, Bangladesh

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### **Network Routing and Switching**

**CCE-415**

## **THEORY ASSIGNMENT – 01**

Short Questions and Answers on Routing

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## 1. Describe the primary functions and features of a router.

A router is a Layer 3 networking device that connects different networks and forwards data packets from one network to another. Its main functions and features are given below.

### Primary Functions:

- **Path Determination:** A router determines the best available path to reach a destination network by using its routing table, routing protocols, and routing metrics.
- **Packet Forwarding:** After determining the appropriate path, the router forwards the packet through the correct outgoing interface toward its destination.
- **Broadcast Control:** Routers normally do not forward Layer 2 broadcast traffic between different interfaces. Therefore, each router interface represents a separate broadcast domain.
- **Connecting Different Networks:** A router allows communication between different IP networks and can connect networks that use different Layer 2 technologies.

### Important Features:

- A router mainly operates at **Layer 3 (Network Layer)** of the OSI model.
- It contains components such as CPU, RAM, ROM, NVRAM, and Flash memory.
- It maintains a **Routing Information Base (RIB)** to store learned routing information.
- It may also use a **Forwarding Information Base (FIB)** for faster packet forwarding.

## 2. Connect devices for a small, routed network.

A small routed network can be created by connecting end devices, switches, and a router in the following way:

1. **Connect PCs to Switches:** Connect each PC to a switch using Ethernet UTP cables. The PCs are normally connected to access ports on the switch.
2. **Connect Switches to the Router:** Connect the switch uplink port to the appropriate router interface, such as `GigabitEthernet0/0/0` or `GigabitEthernet0/0/1`.
3. **Configure the End Devices:** Each PC should be configured with:
  - An IP address

- A subnet mask
- A default gateway

The default gateway should be the IP address of the router interface connected to that particular network.

4. **Console Connection:** During the initial configuration, an administrator can connect a PC to the router's console port using a console cable.

### 3. Configure basic settings on a router to route between two directly-connected networks, using CLI.

Suppose the router connects two LANs. The first network is 192.168.10.0/24 and the second network is 192.168.20.0/24. The router can be configured as follows:

```
Router> enable
Router# configure terminal

! Configure the first LAN
Router(config)# interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

! Configure the second LAN
Router(config)# interface GigabitEthernet0/0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit

! Finish configuration and save it
Router(config)# end
Router# copy running-config startup-config
```

After configuring both interfaces and bringing them up, the router automatically knows both networks because they are directly connected.

### 4. Verify connectivity between two networks that are directly connected to a router.

Connectivity between the two networks can be verified using several commands and tests.

### 1. Check Router Interfaces:

Use the following command:

```
show ip interface brief
```

This command shows the status of all router interfaces. The required interfaces should normally show up/up.

### 2. Check the Routing Table:

Use:

```
show ip route
```

The routing table should contain the two directly connected networks with the code C.

### 3. Test Connectivity Using Ping:

From a PC in the first network, ping a PC in the second network:

```
ping 192.168.20.10
```

A successful reply indicates that the two networks can communicate.

### 4. Use Traceroute:

On a Windows PC, the following command can be used:

```
tracert 192.168.20.10
```

The router should appear as an intermediate hop between the two networks.

### 5. Check ARP Information:

The router's ARP table can be checked using:

```
show ip arp
```

This helps verify the mapping between IP addresses and MAC addresses.

## 5. Explain the encapsulation and de-encapsulation process used by routers when switching packets between interfaces.

When a router receives a packet from one network and forwards it to another network, the Layer 2 frame is removed and a new Layer 2 frame is created.

The process can be explained in four main steps:

#### 1. Receiving and De-encapsulation:

The router receives a Layer 2 frame through its incoming interface. It checks the frame and removes the Layer 2 header and trailer to obtain the IP packet.

## **2. Processing the IP Packet:**

The router examines the destination IP address and checks its routing table to determine where the packet should be forwarded. The router also decreases the TTL value by one.

## **3. Finding the Next Hop:**

After selecting the outgoing interface, the router determines the MAC address of the next-hop device. For IPv4 Ethernet networks, ARP can be used for this purpose.

## **4. Re-encapsulation and Forwarding:**

The router places the IP packet inside a new Layer 2 frame and sends it through the outgoing interface.

- **Source MAC:** MAC address of the router's outgoing interface.
- **Destination MAC:** MAC address of the next-hop device or destination host.

Therefore, the IP packet generally remains the same while the Layer 2 frame is changed at every router hop.

# **6. Explain the path determination function of a router.**

Path determination is the process by which a router selects the best route to a destination from the available routes.

The router mainly considers the following factors:

## **1. Longest Prefix Match (LPM):**

The router first selects the route that provides the most specific match for the destination IP address. For example, a /26 route is more specific than a /24 route.

## **2. Administrative Distance (AD):**

If the same destination network is learned from different routing sources, the route with the lower Administrative Distance is preferred.

For example:

- Connected = 0
- Static = 1

- OSPF = 110
- RIP = 120

### 3. Routing Metric:

If multiple routes are learned from the same routing protocol, the protocol uses its metric to select the best path. For example, RIP uses hop count, while OSPF uses cost.

### 4. Equal-Cost Multi-Path (ECMP):

When multiple paths have the same cost, a router may use multiple paths for load balancing.

## 7. Explain routing table entries for directly connected networks.

When a router interface is configured with an IP address and becomes active, the router normally creates two types of routing entries.

For example:

|   |                                                             |
|---|-------------------------------------------------------------|
| C | 192.168.10.0/24 is directly connected, GigabitEthernet0/0/0 |
| L | 192.168.10.1/32 is directly connected, GigabitEthernet0/0/0 |

Here:

- **C – Connected Route:**

The C entry represents the complete network connected to the router interface. It is used when the destination belongs to that directly connected subnet.

- **L – Local Route:**

The L entry represents the exact IP address configured on the router interface. It uses a /32 host route and helps the router recognize packets that are addressed directly to itself.

Thus, the connected route represents the network, while the local route represents the router's own interface IP address.

## 8. Explain how a router builds a routing table of directly connected networks.

A router automatically adds directly connected networks to its routing table when an interface is correctly configured and active.

The process is as follows:

1. **Configure an IP Address:**

An IP address and subnet mask are assigned to the router interface.

2. **Activate the Interface:**

The `no shutdown` command is used to enable the interface.

3. **Check the Interface Status:**

The physical and data-link layers must become active. The interface should normally show an `up/up` status.

4. **Add the Routes:**

Once the interface is active, the router automatically adds:

- A connected (C) route for the network.
- A local (L) route for the interface's own IP address.

Directly connected routes have an Administrative Distance of **0**. If the interface goes down, the corresponding connected and local routes are removed from the routing table.

## 9. Explain how a router builds a routing table using static routes.

A static route is manually configured by a network administrator. Unlike dynamic routes, static routes do not require a routing protocol to learn the destination network.

The general syntax is:

```
ip route <destination-network> <subnet-mask>  
{next-hop-ip | exit-interface} [AD]
```

Some common types of static routes are:

- **Next-Hop Static Route:**

The administrator specifies the IP address of the next router.

```
ip route 10.1.1.0 255.255.255.0 192.168.12.2
```

- **Exit-Interface Static Route:**

The outgoing interface is specified directly.

```
ip route 10.1.1.0 255.255.255.0 Serial0/1/0
```

- **Default Static Route:**

A default route is used when no more specific route is available.

```
ip route 0.0.0.0 0.0.0.0 192.168.12.2
```

This route is commonly called the **gateway of last resort**.

- **Floating Static Route:**

A floating static route is configured with a higher Administrative Distance so that it can be used as a backup route.

By default, a static route has an Administrative Distance of **1** and is represented by the code **S** in the routing table.

## 10. Explain how a router builds a routing table using a dynamic routing protocol.

Dynamic routing protocols allow routers to automatically learn routes from neighboring routers. This reduces the need for manually configured routes.

The process generally occurs as follows:

1. **Neighbor Discovery:**

Routers discover neighboring routers and establish relationships using routing protocol messages such as Hello packets.

2. **Exchange Routing Information:**

Routers exchange information about available networks, paths, and other routing details. For example, OSPF routers exchange Link-State Advertisements (LSAs).

3. **Calculate the Best Path:**

Each routing protocol uses its own algorithm to select the best path. Examples include:

- OSPF — Dijkstra's Shortest Path First (SPF)
- RIP — Bellman-Ford based distance-vector algorithm
- EIGRP — DUAL

#### 4. **Install the Best Route:**

The selected routes are placed into the routing table. Common routing table codes include:

- O for OSPF
- D for EIGRP
- R for RIP

#### 5. **Maintain and Update Routes:**

If a network link fails or the topology changes, the routing protocol shares the necessary information with other routers and calculates a new best path.

This automatic process of updating routing information and adapting to topology changes is known as **convergence**.

— *End of Assignment* —